

A regular realization of the Suzuki group $Sz(8)$ over $\mathbb{Q}$

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Abstract

We realize the Suzuki group $Sz(8)$ regularly over $\mathbb{Q}$ by an explicit degree-65 family over $\mathbb{Q}(s)$, and give a monic integral degree-65 polynomial over $\mathbb{Q}$ with this Galois group and 43-digit coefficients. Every three-point regular realization of $\text{Aut}(Sz(8)) = Sz(8) \rtimes C_3$ over $\mathbb{Q}(t)$ has a rational cyclic cubic quotient: it is ramified at exactly two geometric points, and its odd degree makes its genus-zero curve rational over $\mathbb{Q}$. We construct such a realization as a genus-six cover of degree 65 branched at $\zeta_3, \zeta_3^2, \infty$, and pull back along an explicit cubic map $t = T(s)$. A monic polynomial $f(X, t)$ of bidegree $(65, 7)$ is certified directly from its coefficients by an exact discriminant computation, a local argument removing 342 apparent branch points, and certified continuation.

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*The computations, verification programs, and text of this paper were produced with extensive use of large language models under the author's direction; the author takes full responsibility for the contents.

1 Introduction

The inverse Galois problem asks whether every finite group occurs as the Galois group of a finite extension of $\mathbb{Q}$. The Suzuki group

$$N = \text{Sz}(8), \quad |N| = 29120 = 2^6 \cdot 5 \cdot 7 \cdot 13,$$

is the smallest of the seven simple groups of order at most 10^8 on Zywina's August 2025 list of unresolved cases [Zyw25]. The case of M_{23} on that list has since been resolved by Huang–Jackson–Lee–Poonen–Pries–Zhang [HJL⁺26]. We construct a regular realization of N , and a specific number-field realization, by computing an auxiliary cover with group

$$G = \text{Aut}(N) = N \rtimes C_3.$$

Throughout, regular means that the algebraic closure of $\mathbb{Q}$ in the Galois extension is $\mathbb{Q}$.

The idea. For a three-point regular G -extension $L/\mathbb{Q}(t)$, the fixed field L^N is cyclic cubic and ramified at exactly two geometric points. Its curve has genus zero, and the degree-three map to $\mathbb{P}^1$ supplies a rational divisor of odd degree, hence a rational point. Thus $L^N = \mathbb{Q}(s)$, and $L/\mathbb{Q}(s)$ realizes N regularly (Corollary 2.3). This reduces the problem to constructing a three-point realization of G and identifying its cubic quotient map. The cover used here has genus 6 in the degree-65 action; its passport has four Nielsen classes, so rigidity alone does not supply the realization.

Let $f(X, t) \in \mathbb{Z}[1/13][X, t]$ be the exact polynomial supplied in `data/f.json`, monic of degree 65 in X and of degree 7 in t . Put

$$T(s) = \frac{s^3 + 6s^2 - 27s + 21}{3(s^3 - 7s^2 + 12s - 5)}. \quad (1)$$

Theorem 1.1. *The splitting field of $f(X, t)$ over $\mathbb{Q}(t)$ is regular with group $\text{Sz}(8) \rtimes C_3$. The splitting field of $f(X, T(s))$ over $\mathbb{Q}(s)$ is regular with group $\text{Sz}(8)$. Moreover,*

$$\text{Gal}(f(X, -7/5)/\mathbb{Q}) \simeq \text{Sz}(8).$$

Here the Galois group of a polynomial means the Galois group of its splitting field.

A smaller presentation is the monic polynomial $F_3(Y, U)$ of bidegree $(65, 7)$ supplied in `data/F3_L3POP4.json`, with $U = 3(t + 1)$. An exact change of generator relates it to f (Section 6).

Corollary 1.2. *The splitting field of $F_3(Y, 3(t + 1))$ over $\mathbb{Q}(t)$ is regular with group $\text{Sz}(8) \rtimes C_3$. Its pullback $F_3(Y, 3(T(s) + 1))$ has regular splitting field with group $\text{Sz}(8)$ over $\mathbb{Q}(s)$. The monic integral polynomial P_1 printed in Appendix A defines the degree-65 field obtained by specializing this cover at $s = 1$; its Galois group over $\mathbb{Q}$ is $\text{Sz}(8)$ and its signature is $(1, 32)$.*

The auxiliary degree-65 cover has genus 6 and branch-cycle types

branch point	ζ_3	ζ_3^2	∞
cycle type	$1^5 3^{20}$	$1^5 3^{20}$	13^5 .

(2)

The cyclic cubic quotient is branched at the two finite branch points. Determining its arithmetic form gives (1). After this base change the degree-65 cover has genus 26, with three geometric branch points of type 13^5 .

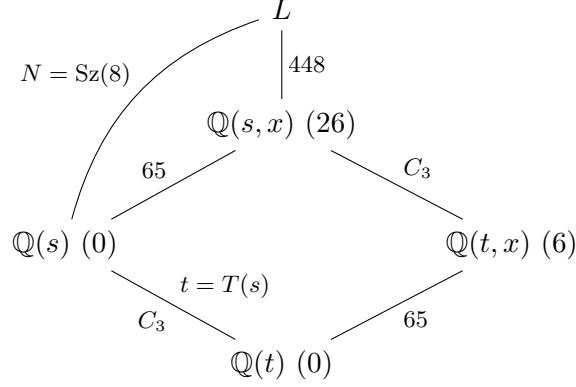


Figure 1: The fields in [Theorem 1.1](#). Here L is the splitting field of $f(X, t)$ over $\mathbb{Q}(t)$, with $\text{Gal}(L/\mathbb{Q}(t)) = G = \text{Sz}(8) \rtimes C_3$, and x is a root of $f(X, t)$. Edge labels are degrees or Galois groups; numbers in parentheses are genera. The fixed field $\mathbb{Q}(s) = L^N$ is rational by [Lemma 2.1](#), and $t = T(s)$ is [\(1\)](#).

Contributions and related work. The main contribution is the explicit regular realization in [Theorem 1.1](#). Two features make the computation practical. Functions with poles supported over infinity give predictable coefficient-degree bounds and much smaller equations ([Section 6](#)). A local discriminant argument removes hundreds of apparent branch points before monodromy is tracked ([Section 3](#)).

Rational-quotient descent has classical precedents discussed in [Section 2](#). Numerical construction of three-point covers follows the triangle-group approach of Klug–Musty–Schiavone–Voight [[KMSV14](#)]; see also the survey [[SV14](#)]. Certified path tracking for Belyi maps has substantial prior work, including Guillemot–Voight’s certification of the Belyi maps in the LMFDB [[GV26](#)]. Our emphasis is the arithmetic application and the concrete bounds for this singular plane model. For the broader arithmetic context of Hurwitz spaces, see Dèbes [[Dèb26](#)].

Proof structure. The proof starts from the exact coefficients of f and is independent of the numerical construction. After the quotient criterion in [Section 2](#), [Section 3](#) establishes geometric monodromy and regularity. [Section 4](#) identifies the cubic map from its fibers at infinity and at $t = 0$, proving the regular $\text{Sz}(8)$ realization. [Section 5](#) completes [Theorem 1.1](#) with a rational specialization. [Section 6](#) then transfers the certificate to F_3 by an exact change of generator and proves [Corollary 1.2](#), including the shorter printed specialization. [Section 7](#) describes discovery, and [Section 8](#) documents the verification programs and their exact inputs.

2 Rational cyclic cubic quotients

The degree-65 permutation representation of $N = \text{Sz}(8)$ extends to $G = N \rtimes C_3$. The element orders in N are 1, 2, 4, 5, 7, 13. Thus order-3 inertia maps nontrivially to G/N , whereas order-13 inertia maps trivially. The passport [\(2\)](#) therefore gives a cyclic cubic quotient ramified at exactly two geometric points.

Lemma 2.1. *Let $L/\mathbb{Q}(t)$ be a regular Galois extension with group G , and let $N \triangleleft G$ with $G/N \simeq C_3$. If $M = L^N$ is ramified over exactly two points of $\mathbb{P}^1(\overline{\mathbb{Q}})$, then $M \simeq \mathbb{Q}(s)$. In particular, $L/\mathbb{Q}(s)$ is a regular Galois extension with group N .*

Proof. Let C be the smooth projective curve with function field M . Its degree-three map to $\mathbb{P}^1$ is totally ramified at each branch point, since a nontrivial inertia subgroup of a cyclic group of prime order is the whole group. Riemann–Hurwitz gives

$$2g(C) - 2 = 3(-2) + 2(3 - 1) = -2.$$

The pullback of any rational point of $\mathbb{P}^1$ is a rational divisor of degree 3 on C . Adding a canonical divisor gives a rational divisor of degree 1, which is linearly equivalent to an effective divisor by Riemann–Roch. Thus C has a rational point and $C \simeq \mathbb{P}_{\mathbb{Q}}^1$. Regularity is inherited from L . $\square$

The branch cycle lemma relates the Galois action on branch points to cyclotomic powers of inertia generators [Fri77]; see also [Völ96, p. 34]. For a constant cyclic group, conjugacy classes are singletons. This gives a useful restriction on the quotient geometry.

Proposition 2.2. *Let $M/\mathbb{Q}(t)$ be a regular Galois extension with constant group C_ℓ , where ℓ is an odd prime, and let r be its number of geometric branch points. Every Galois orbit of branch points has size divisible by $\ell - 1$, and*

$$g(M) = \frac{(\ell - 1)(r - 2)}{2} \geq \frac{(\ell - 1)(\ell - 3)}{2}.$$

In particular, for $\ell = 3$ one has $r = 2k$ and $g(M) = 2k - 2$, with $k \geq 1$.

Proof. Fix a branch point a and its nonidentity inertia generator g_a . If $\tau \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ fixes a , the branch cycle lemma gives $g_a^{\chi_\ell(\tau)} = g_a$; with the inverse branch-cycle convention the exponent is $\chi_\ell(\tau)^{-1}$ instead. Here χ_ℓ is the cyclotomic character modulo ℓ . Since g_a has order ℓ , either equality forces $\chi_\ell(\tau) = 1$, and hence $\mathbb{Q}(\zeta_\ell) \subseteq \mathbb{Q}(a)$. Thus $\ell - 1$ divides the size of each branch-point orbit. The extension cannot be geometrically unramified, since $\mathbb{P}_{\mathbb{C}}^1$ has no connected nontrivial unramified cover. Hence $r \geq \ell - 1$. Every branch point is totally ramified, so Riemann–Hurwitz gives $2g(M) - 2 = -2\ell + r(\ell - 1)$, as asserted. $\square$

Corollary 2.3. *For a regular G -extension $L/\mathbb{Q}(t)$ with $G/N \simeq C_3$, the fixed field L^N is rational over $\mathbb{Q}$ if and only if exactly two branch cycles lie outside N . In particular, every three-point regular realization of $\text{Aut}(\text{Sz}(8))$ over $\mathbb{Q}(t)$ yields a regular realization of $\text{Sz}(8)$ over $\mathbb{Q}(s)$ by a cyclic cubic base change.*

Proof. A branch point survives in L^N precisely when its inertia generator lies outside N . The proposition shows that the quotient has genus zero exactly when two branch points survive, and Lemma 2.1 then gives rationality. For a three-point cover, the number surviving is positive, even, and at most three, hence two. $\square$

The next cyclic cubic possibility has genus 2. For $\ell = 5$ the lower bound is already 4; in particular, the quotient by $\text{Sz}(32)$ in a regular realization of $\text{Aut}(\text{Sz}(32)) = \text{Sz}(32) \rtimes C_5$ cannot be rational. For index 2, genus zero alone need not imply rationality: a rational point on the quotient conic is additional input. Serre’s double group trick supplies such a point when the branch points are rational [Ser08, Lemma 4.5.1]. Malle’s rational translates [Mal91] and König’s index-two descent [Kön17, Proposition 3] are related precedents; outer automorphisms also appear in [Mal92]. In contrast, Zywin’s realization of $\text{PSL}_3(\mathbb{F}_4)$ uses rational points on a genus-three quotient [Zyw25, §1].

Remark 2.4. The rational curve in [Lemma 2.1](#) is a quotient of the Galois closure. The degree-65 source curve has genus 6 before base change and genus 26 afterwards. The quotient criterion therefore remains useful even when the degree-65 source has positive genus.

For the construction we restrict the two outer inertia orders to 3. The remaining inertia order is then 5, 7, or 13, giving source genera 2, 3, and 6, respectively ([Section 7.1](#)). The genus-two construction produced a cover over a cubic number field; the genus-six construction produced the cover over $\mathbb{Q}$ used here. We now certify its polynomial, then identify the arithmetic form of the cubic quotient.

3 An independent certificate for the polynomial

Starting from the coefficients of f , we determine its discriminant, remove the apparent branch points, and certify two monodromy permutations. Their group then gives both geometric monodromy and regularity. Put $q(t) = t^2 + t + 1$.

Proposition 3.1 (Computational certificate). *The following statements hold for f .*

(i) *Its exact X -discriminant is*

$$\text{disc}_X(f) = c q(t)^{40} S(t)^2, \tag{3}$$

where $c \in \mathbb{Q}^\times$, $S \in \mathbb{Q}[t]$ is monic and squarefree of degree 342, and $\gcd(S, q) = 1$.

- (ii) *At each root a of S , $f(X, a)$ has exactly one repeated root, and that root has multiplicity exactly 2.*
- (iii) *Over $\mathbb{Q}(\zeta_3)$ one has $f(X, \zeta_3) = f_5(X) f_{20}(X)^3$, with distinct irreducible factors of the indicated degrees, and $\text{Res}_X(f_{20}, f_t(X, \zeta_3)) \neq 0$.*
- (iv) *Continuation from $t = 0$ along standard meridians around ζ_3 and ζ_3^2 gives permutations σ_1, σ_2 of type $1^5 3^{20}$. With permutations composed from left to right, $\sigma_\infty = (\sigma_1 \sigma_2)^{-1}$ has type 13^5 , and they generate a transitive primitive group of order 87360.*

The following computations establish the proposition; the programs and exact outputs are documented in [Section 8](#).

3.1 The exact discriminant

The general discriminant degree bound is $\deg_t \text{disc}_X(f) \leq 2(65 - 1)7 = 896$. Exact evaluations at 897 distinct rational values of t and exact interpolation therefore determine the discriminant. Additional evaluations check the interpolation. Exact division and gcd computations give (3), squarefreeness, and coprimality. The resulting discriminant has degree $764 = 80 + 2 \cdot 342$.

3.2 Nodes of the plane model

Lemma 3.2. *Let $F(X, t)$ be monic in X with coefficients holomorphic near $t = a$. Suppose $\text{ord}_{t=a} \text{disc}_X(F) = 2$ and $F(X, a)$ has exactly one repeated root, of multiplicity 2. Then the normalization of $F(X, t) = 0$ is unramified above a .*

Proof. Separate the simple roots holomorphically. Weierstrass preparation at the double root reduces the remaining two roots to a quadratic factor $X^2 + u(t)X + v(t)$, after translating X . Its discriminant has order exactly 2, because the discriminants and mutual resultants of the other factors are units. Thus $u(t)^2 - 4v(t) = (t - a)^2 w(t)$ with $w(a) \neq 0$. A local holomorphic square root of w gives the two roots

$$\frac{-u(t) \pm (t - a)\sqrt{w(t)}}{2}.$$

They are single-valued holomorphic functions of t . $\square$

There is an exact modular check of [Proposition 3.1\(ii\)](#). Let $B(t)$ be the first principal subresultant coefficient of f and f_X , considered as polynomials in X . At a root a of S , their resultant vanishes, and $B(a) \neq 0$ is equivalent to $\deg \gcd(f(X, a), f_X(X, a)) = 1$. In characteristic zero this degree is $\sum_j (m_j - 1)$, where m_j are the root multiplicities. Degree one therefore means exactly one double root, with all other roots simple.

It suffices to prove $\text{Res}_t(S, B) \neq 0$ modulo one suitable prime. At $p = 1000003$, exact discriminant interpolation modulo p gives degree 764 and the shape $\bar{c}\bar{q}^{40}\bar{S}^2$, with $\bar{S}$ monic and squarefree of degree 342 and coprime to $\bar{q}$. The full-degree reduction ensures that the characteristic-zero monic factor S is p -integral and reduces to this $\bar{S}$. Its irreducible factors have degrees

$$1, 3, 5, 14, 78, 84, 157.$$

For every factor P , the gcd of $\bar{f}$ and $\bar{f}_X$ over $\mathbb{F}_p[t]/(P)$ has degree one. Subresultants commute with these specializations because the leading coefficients 1 and 65 remain units. Hence $\bar{B}$ is nonzero at every root of $\bar{S}$, proving the required nonvanishing in characteristic zero. The independent verifier `scripts/check_nodes_modp.py` runs this finite-field calculation from the coefficients of f ; it uses the exact discriminant identity [\(3\)](#) to draw its characteristic-zero conclusion.

The certificate also contains an interval proof. For each root of S , it first certifies a parameter rectangle T containing exactly one root by interval Newton on S . The 342 rectangles are pairwise disjoint, hence account for all roots of S . It then isolates all 64 roots of $f_X(X, t)$ uniformly for $t \in T$ in disjoint rectangles. The value of f excludes zero on all but one of these rectangles, and f_{XX} excludes zero on the remaining rectangle. The vanishing of the discriminant supplies existence of a repeated root; these tests establish its uniqueness and multiplicity.

The enclosure conventions are recorded in [Section 8.2](#). Together with [Lemma 3.2](#), either test proves that the normalized cover has no finite branch point outside $\{\zeta_3, \zeta_3^2\}$.

At a root of f_{20} over $t = \zeta_3$, the nonvanishing of f_t in [Proposition 3.1\(iii\)](#) makes the plane curve smooth and expresses $t - \zeta_3$ as a function with a zero of order 3 in a local X -coordinate. The twenty such points account for discriminant order $20(3 - 1) = 40$. The five simple roots are unramified. Complex conjugation gives the same description over ζ_3^2 .

3.3 Certified continuation

The base point is $t = 0$. A *standard meridian* goes from the base point along a tail to a small neighborhood of one branch point, makes one counter-clockwise circuit, and returns along the same tail. Here the tails are straight, with entry point $z - \varepsilon z/|z|$ for $z = \zeta_3, \zeta_3^2$; the circuits are inscribed polygons in disjoint small neighborhoods of the branch points. Path parameters are recorded in [Section 8.2](#). The two meridians wind once about their respective branch point and zero times about the other, and form a generating system for the thrice-punctured sphere. Apparent branch points have already been removed by [Lemma 3.2](#).

At a parameter value t_0 , isolate the roots of $f(X, t_0)$ in rectangles with centers c_i and enclosing Euclidean radii r_i . Choose disjoint discs $|X - c_i| < R_i$ containing those rectangles. On the i th boundary circle, the monic factorization gives

$$|f(X, t_0)| \geq (R_i - r_i) \prod_{\ell \neq i} (|c_i - c_\ell| - R_i - r_\ell). \quad (4)$$

For each $1 \leq j \leq 7$, isolate the roots of $\partial_t^j f(X, t_0)$. If its leading coefficient is a_j and its roots have centers $b_{j\ell}$, enclosing radii $\rho_{j\ell}$, and multiplicities $m_{j\ell}$, then

$$|\partial_t^j f(X, t_0)| \leq |a_j| \prod_{\ell} (|c_i - b_{j\ell}| + R_i + \rho_{j\ell})^{m_{j\ell}}.$$

The Taylor expansion in t terminates at degree 7. Bounding its j th term by $h^j/j!$ times this product gives a Rouché test valid for every $|t - t_0| \leq h$. For each candidate radius, bisection selects h such that the sum of the seven upper bounds is strictly below (4); the common step is the minimum over all roots. Factoring the t -derivatives avoids severe cancellation losses in bounds obtained from absolute values of coefficients.

Each accepted endpoint is checked in ball arithmetic to lie within the certified parameter displacement. Isolated endpoint roots are matched by containment in the preceding discs. Thus labels persist along the whole path. The resulting permutations are supplied in `data/monodromy.json`.

There is a useful symmetry check. Let c be complex conjugation on the isolated roots of $f(X, 0)$. The two meridians are conjugate up to homotopy with reversed orientation, so

$$\sigma_2 = c \sigma_1^{-1} c.$$

Matching conjugate root enclosures gives an involution of type $1\,2^{32}$, and the permutations obtained by tracking both loops satisfy this identity exactly.

3.4 Geometric group and regularity

By [Lemma 3.2](#), the two meridians generate the fundamental group needed for the normalized cover. Transitivity of their action proves connectedness. The GAP primitive-group library has 13 groups of degree 65; exactly one has order 87360, namely $G = \text{Sz}(8) \rtimes C_3$. This identifies the geometric monodromy in [Proposition 3.1\(iv\)](#).

The arithmetic monodromy contains G normally and is contained in $N_{S_{65}}(G)$. This normalizer is G itself. One can see this without a second search: the simple derived subgroup N is characteristic in G , the centralizer of its nonregular doubly transitive action is trivial, and $\text{Aut}(N) = G$. Conjugation therefore embeds the normalizer in G , while it already contains G . Arithmetic and geometric monodromy coincide, proving the first assertion of [Theorem 1.1](#).

3.5 Two genus checks from the plane equation

Write $f = \sum_{j=0}^7 C_j(X) t^j$. Exact factorization gives $C_7 = aH(X)^{13}$ with H an irreducible separable quartic, and $\gcd(H, C_6) = 1$. In the chart $u = 1/t$, the equation $u^7 f(X, 1/u) = 0$ is consequently smooth at the four roots of H on $u = 0$, with u vanishing to order 13. At $(X, t) = (\infty, \infty)$, put $v = 1/X$. The lowest terms of $v^{65} u^7 f(1/v, 1/u)$ for weights $\text{wt}(v) = 7$, $\text{wt}(u) = 13$ are $u^7 + av^{13}$. Since 7 and 13 are coprime, this is one branch with $\delta = (7 - 1)(13 - 1)/2 = 36$. These coefficient checks are included in `scripts/check_review_evidence.py`.

The 342 points in [Lemma 3.2](#) are ordinary nodes: the two local roots have distinct derivatives in t . All remaining points are smooth by the preceding local checks. The closure in $\mathbb{P}^1 \times \mathbb{P}^1$ has arithmetic genus $(65 - 1)(7 - 1) = 384$, so its geometric genus is

$$384 - 342 - 36 = 6.$$

There is also a discriminant-degree check. The thirteen unbounded roots grow like $t^{7/13}$; the other fifty-two approach the four distinct roots of H , with differences within each group of thirteen of order $t^{-1/13}$. Thus the squared pairwise differences give

$$\deg_t \operatorname{disc}_X(f) = 2 \binom{13}{2} \frac{7}{13} + 2(13 \cdot 52) \frac{7}{13} - 2 \left(4 \binom{13}{2} \right) \frac{1}{13} = 84 + 728 - 48 = 764.$$

Both checks agree with the certified discriminant and branch cycles.

4 Identifying the cyclic cubic quotient

Let L be the splitting field of f over $\mathbb{Q}(t)$ and put $M = L^N$. By [Proposition 3.1](#), inertia at the two finite branch points has order 3, and inertia at infinity has order 13. [Lemma 2.1](#) gives $M = \mathbb{Q}(s)$. To obtain an explicit polynomial with group N , we determine t as a rational function of s .

4.1 The fiber at infinity

Fix a place of L above ∞ , with inertia I and decomposition group D . Tame ramification of index 13 puts μ_{13} in its residue field k . Since the base place is rational,

$$D/I = \operatorname{Gal}(k/\mathbb{Q}) \twoheadrightarrow \operatorname{Gal}(\mathbb{Q}(\zeta_{13})/\mathbb{Q}) \simeq C_{12}.$$

On the other hand, $D \subseteq N_G(I)$ and the latter has order 156. Thus $D = N_G(I)$ and $k = \mathbb{Q}(\zeta_{13})$. Moreover, $D \cap N$ has order 52. The residue field in the cubic quotient is therefore the unique cubic subfield of $\mathbb{Q}(\zeta_{13})$, represented by

$$z^3 + z^2 - 4z + 1.$$

The orbits of D on the 65 letters have lengths 13 and 52. The inertia group has five orbits of length 13; D/I fixes one of these orbits and permutes the other four transitively. The corresponding closed points therefore have degrees $13/13 = 1$ and $52/13 = 4$. Denote these points by P_0 and P_4 , respectively. On the degree-65 source curve the pole divisor is therefore

$$(t)_\infty = 13(P_0 + P_4). \tag{5}$$

The residue field at P_4 is the unique quartic subfield of $\mathbb{Q}(\zeta_{13})$. Geometrically, P_4 becomes four points, each of ramification index 13; together they account for the 52 sheets, while P_0 accounts for the other 13. See also [Figure 1](#).

4.2 Two twists and one distinguishing prime

Consider the reference cyclic cubic cover

$$u^3 - 3tu^2 - (3t + 3)u - 1 = 0. \tag{6}$$

Its discriminant is $81(t^2 + t + 1)^2$, its deck group is constant C_3 , and its fiber at infinity consists of three rational points. Over $\overline{\mathbb{Q}}$, a cyclic cubic cover with these two branch points is unique up to isomorphism over the base. Its $\mathbb{Q}$ -forms are consequently classified by

$$H^1(\mathbb{Q}, C_3) = \text{Hom}_{\text{cont}}(\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), C_3),$$

because the deck group is constant. The split fiber over infinity is a C_3 -torsor. Twisting by a character χ changes its Galois action from the trivial action to translation by χ ; when χ is surjective, its residue field is therefore the fixed field of $\ker(\chi)$. Requiring this field to be the conductor-13 cubic field K leaves precisely the two isomorphisms $\text{Gal}(K/\mathbb{Q}) \simeq C_3$, which are inverse to one another. One can construct their twists over the cyclic cubic field $K = \mathbb{Q}[z]/(z^3 + z^2 - 4z + 1)$: if τ generates $\text{Gal}(K/\mathbb{Q})$ and $\rho(u) = -1/(u + 1)$ generates the deck group of (6), descend $K(u)$ by the semilinear actions (τ, ρ) and (τ, ρ^{-1}) . Rational coordinates on the two fixed fields give the following maps, whose discriminants and fibers provide exact checks of the construction:

$$T_1(s) = \frac{-4s^3 + 6s^2 + 36s - 6}{3(s^3 + 5s^2 - 9s - 5)}, \quad (7)$$

$$T_2(s) = \frac{s^3 + 6s^2 - 27s + 21}{3(s^3 - 7s^2 + 12s - 5)}. \quad (8)$$

Writing $T_i = n_i/d_i$ with the displayed numerators and denominators, the exact identities

$$\text{disc}_s(n_1 - td_1) = 936^2(t^2 + t + 1)^2, \quad \text{disc}_s(n_2 - td_2) = 117^2(t^2 + t + 1)^2$$

verify their branch loci and cyclicity. Both denominator fields are isomorphic to the conductor-13 field above. Their fibers at $t = 0$ are the distinct cyclic cubic fields represented by $z^3 - 39z - 26$ and $z^3 - 39z - 91$, respectively. Both have field discriminant 117^2 , hence conductor $117 = 9 \cdot 13$: they combine the conductor-9 fiber of the reference cover at $t = 0$ with the two conductor-13 characters. The accompanying program `scripts/check_cubic_maps.py` checks these discriminants and field isomorphisms exactly with PARI/GP.

At a prime of separable reduction, the fiber of M over $t = 0$ splits if and only if the Frobenius element of $f(X, 0)$ belongs to N . Every element of $G \setminus N$ has order divisible by 3, and N has no element of order 3. Thus membership is read from the factor degrees: the Frobenius belongs to N precisely when none is divisible by 3.

At $p = 29$, the factorization type of $f(X, 0)$ is $1\,2^2\,6^{10}$, so Frobenius lies outside N . The cubic for T_1 splits modulo 29, whereas the cubic for T_2 does not. Since the two twists exhaust the possibilities, this single splitting test identifies M with $\mathbb{Q}(s)$ via $t = T_2(s)$.

The equality $L^N = \mathbb{Q}(s)$ proves that $L/\mathbb{Q}(s)$ is regular with group N . Since N is transitive on the 65 roots, $f(X, T_2(s))$ is irreducible. For a polynomial with coefficients in $\mathbb{Q}[s]$ we clear denominators:

$$g(X, s) = (3(s^3 - 7s^2 + 12s - 5))^7 f(X, T_2(s)). \quad (9)$$

The polynomial g has bidegree $(65, 21)$ and the same splitting field. Its leading coefficient in X is the displayed seventh power; the monic polynomial is $f(X, T_2(s)) \in \mathbb{Q}(s)[X]$.

5 A number field with group $\text{Sz}(8)$

We complete [Theorem 1.1](#) with a rational fiber of the regular family. Specialize $s = 0$ in (8), so $t = -7/5$, and put $h(X) = f(X, -7/5)$. This is a monic degree-65 polynomial in $\mathbb{Z}[1/65][X]$. Its reductions at 11 and 31 are separable and have the following factorizations by degree:

Prime	Factorization type	Frobenius order
11	$1^2 7^9$	7
31	13^5	13

Separability at either prime proves separability over $\mathbb{Q}$. The map T has no pole at $s = 0$, and h is separable, so this is an unramified specialization of the generic splitting field. Its Galois group H embeds in N . The two reductions give elements of orders 7 and 13 in H , so 91 divides $|H|$.

The maximal subgroups of N have orders

$$448, \quad 52, \quad 20, \quad 14.$$

None is divisible by 91. Hence $H = N$, proving the final assertion of [Theorem 1.1](#).

The degree-65 field $\mathbb{Q}[X]/(h)$ has signature $(1, 32)$. Indeed, complex conjugation on the roots over the real base point $t = 0$ has type $1 2^{32}$, as certified by conjugate root enclosures. There is no finite real branch point on the normalization, so this action has the same cycle type along the real t -line. In particular, every separable real specialization of f has exactly one real root. The verifier also checks the signature directly at $t = -7/5$.

For $s = 0$, the monic integral polynomial P in `data/P_number_field.json` has degree 65 and height 68 digits. It defines the same field as $h(X) = f(X, -7/5)$, whose primitive integer height is 252 digits. The file gives a rational polynomial A taking its root to a root of h ; exact composition modulo the defining polynomial and irreducibility verify the field isomorphism. An inverse is also recorded.

Hilbert irreducibility also supplies infinitely many $\text{Sz}(8)$ extensions from the regular family. The specialization above gives a specific example without requiring an ineffective choice of parameter.

6 Smaller presentations and a printed specialization

A generator with poles distributed over the fiber at infinity gives a much smaller coefficient height for an equation of the same bidegree. We first explain the degree bound, then prove the change of generator by exact resultant identities and give a compact integral specialization. For a nonzero rational polynomial, its *primitive integer height* is the maximum number of decimal digits of the absolute values of its coefficients after clearing denominators and dividing out their common integer factor. All heights below use this convention.

6.1 Coefficient degrees from pole orders

Lemma 6.1. *Suppose $t : Z \rightarrow \mathbb{P}^1$ has degree n , and x generates $\mathbb{Q}(Z)/\mathbb{Q}(t)$ with all its poles above ∞ . At each geometric point P above ∞ , let e_P be the ramification index and let k_P be the pole order of x , or 0 if x is regular there. Form a list $\lambda_1 \geq \dots \geq \lambda_n$ by repeating k_P/e_P exactly e_P times for each P . Then the minimal polynomial*

$$F(X, t) = X^n + \sum_{i=1}^n c_i(t) X^{n-i}$$

has $c_i(t) \in \mathbb{Q}[t]$ and

$$\deg_t c_i \leq \lfloor \lambda_1 + \dots + \lambda_i \rfloor.$$

Proof. The function x is regular above the affine t -line, so is integral over $\mathbb{Q}[t]$. Its minimal polynomial therefore has coefficients in $\mathbb{Q}[t]$. At infinity, each of the e_P local sheets associated with P grows at most like t^{k_P/e_P} . Every term in the i th elementary symmetric function is therefore bounded by a constant times $|t|^{\lambda_1 + \dots + \lambda_i}$. Since c_i is a polynomial, its degree satisfies the asserted bound. $\square$

For a generator with pole divisor $3P_0 + P_4$ as in (5), the list has thirteen entries $3/13$ and fifty-two entries $1/13$, so

$$\deg_t c_i \leq \left\lfloor \frac{3 \min(i, 13) + \max(i - 13, 0)}{13} \right\rfloor \leq 7.$$

Thus such a generator gives an equation of t -degree at most 7. The exact coefficients of F_3 attain this bound. The affine substitution $U = 3(t + 1)$ preserves the degree.

6.2 The smaller family and its exact bridge

The polynomial $F_3(Y, U)$ has 318 nonzero terms and primitive integer height 47 digits, compared with 249 digits for f , although F_3 is the denser polynomial. In the coordinate $U = 3(t + 1)$ its leading terms are

$$\begin{aligned} F_3(Y, U) = & Y^{65} + 78Y^{64} + 2652Y^{63} + 48880Y^{62} + 457290Y^{61} \\ & + (296636 - 13312U)Y^{60} + \dots \end{aligned} \tag{10}$$

The full file `data/F3_L3P0P4.json` uses the exponent-pair encoding in [Section 8.3](#). Its generator comes from the two-dimensional space $L(3P_0 + P_4)$ in the discovery model. The following criterion proves its relation to f .

Lemma 6.2 (Exact change of generator). *Let K be a field of characteristic zero, and let $F(X), P(Y)$ be monic irreducible polynomials of the same degree in $K[X]$ and $K[Y]$. For $\Psi \in K[X, Y]$, put*

$$A(X) = \text{Res}_Y(P(Y), \Psi(X, Y)), \quad B(Y) = \text{Res}_X(F(X), \Psi(X, Y)).$$

If $F \mid A$, $F^2 \nmid A$, $P \mid B$, and $P^2 \nmid B$, then $\Psi(\alpha, \beta) = 0$ gives a Galois-equivariant bijection between the roots of F and P . Corresponding roots generate the same extension of K , and the two splitting fields coincide.

Proof. Since P is monic, $A(X) = \prod_{P(\beta)=0} \Psi(X, \beta)$. Irreducibility and separability of F , together with its multiplicity one in A , show that each root α is a simple root of A . Exactly one factor of the product therefore vanishes at α . The same argument with B gives uniqueness in the reverse direction. The relation is defined over K , so the bijection commutes with every automorphism of an algebraic closure of K . Corresponding roots have equal stabilizers, proving the field assertions. $\square$

Apply the lemma over $K = \mathbb{Q}(t)$ with $F = f(X, t)$ and $P = F_3(Y, 3(t+1))$. The supplied bridge file records a relation Ψ of bidegree $(7, 7)$ and the affine change of generator used to express it. The verifier checks the relation with that change applied; the public F_3 agrees after the parameter substitution $U = 3(t+1)$. Irreducibility of P is checked at $t = 0$. The two resultant remainders have t -degree at most 294 and 154: division by a monic polynomial preserves the weights supplied by the pole bounds. Their exact vanishing at 295 and 155 rational values, respectively, proves the two divisibilities. Coprimality of each quotient with its divisor at $t = 0$ proves multiplicity one.

It follows that F_3 has the same splitting field as f over $\mathbb{Q}(t)$. Writing $d(s) = 3(s^3 - 7s^2 + 12s - 5)$, its polynomial pullback is

$$G_3(Y, s) = d(s)^7 F_3(Y, 3(T(s) + 1)). \quad (11)$$

It has bidegree $(65, 21)$. The stored pullback uses $s \mapsto s + 1$ and has primitive integer height 52 digits; its exact substitution identity is checked separately. This proves the generic assertions of [Corollary 1.2](#).

6.3 A printed specialization at $s = 1$

For $s = 1$, so $t = 1/3$ and $U = 4$, the polynomial $F_3(Y, 4)$ has height 48 digits. Its separable reductions have types 13^5 at 11 and $1^2 7^9$ at 17. This is a separable specialization of the regular N -cover defined by $F_3(Y, 3(T(s)+1))$. Its group embeds in N , and the maximal-subgroup argument of [Section 5](#) gives Galois group $\text{Sz}(8)$.

The monic polynomial P_1 in `data/P_s1_number_field.json`, printed in [Appendix A](#), has height 43 digits, compared with 68 digits for the reduced generator at $s = 0$. The file supplies $A_1 \in \mathbb{Q}[X]$ with $F_3(A_1(X), 4) \equiv 0 \pmod{P_1(X)}$. Exact irreducibility and this identity show that P_1 defines the same specialization field. Certified conjugate root matching gives its signature $(1, 32)$, completing [Corollary 1.2](#). The program `scripts/check_review_evidence.py` checks these statements and compares all printed coefficients with the file.

7 Discovery

Chen's interpretation of noncongruence modular curves as Hurwitz spaces was the starting point [[Che25](#), §5.4]. On his distinguished component, one positive-rank rational fiber would suffice [[Che25](#), Proposition 5.6 and the following remark], but our computation found none. We instead searched for three-point $\text{Aut}(\text{Sz}(8))$ -covers whose cyclic cubic quotient could yield $\text{Sz}(8)$ after base change; an order-three symmetry of a $(5, 5, 5)$ cover suggested this route.

7.1 The three passports with outer inertia of order three

For a three-point regular G -cover, the two outer branch points form one Galois orbit. The proof of [Proposition 2.2](#) shows that their common residue field is exactly $\mathbb{Q}(\zeta_3)$. The third branch point is rational. A fractional linear transformation over $\mathbb{Q}$ places the third point at ∞ and the conjugate pair at ζ_3, ζ_3^2 .

We restrict the two outer inertia orders to 3. There is one order-3 class D in each outer coset, and the two classes are inverses. The class in the rational branch slot must be rational in G . The order-7 classes of N fuse in G , as do the order-13 classes; together with the order-5 class they give the following possibilities. The two order-4 classes are inverse and remain distinct, so neither is rational. The involution class, of type $1\,2^{32}$, would give the impossible genus $1 - 65 + (32 + 40 + 40)/2 = -8$. These observations make the table exhaustive under the stated order-3 restriction.

Order in the rational slot	Cycle type	Source genus	Nielsen classes
5	5^{13}	2	3
7	$1^2 7^9$	3	6
13	13^5	6	4

The other two cycles have type $1^5 3^{20}$. For example, the last row has genus $1 - 65 + (60 + 40 + 40)/2 = 6$.

The counts are for generating product-one triples in (C, D, D^{-1}) with ordered branch slots, modulo simultaneous conjugation by G . Fixing the first entry $c \in C$ leaves respectively 45, 42, and 52 choices of the second entry that generate G and give the required third entry. The centralizers of c have orders 15, 7, and 13. Their action is free because a generating triple has centralizer $Z(G) = 1$; division gives 3, 6, and 4. Thus the $(13, 3, 3)$ passport is not rigid. Complex conjugation fixes two of its four classes and interchanges the other two. Reality alone does not determine which classes descend to $\mathbb{Q}$; the polynomial certificate proves descent for the class used here.

The order-5 construction supplied a genus-two cover over $\mathbb{Q}(\theta)$ with $\theta^3 - \theta^2 + 2\theta + 2 = 0$. The order-13 row supplied the rational cover.

7.2 Numerical construction and arithmetic coordinates

The triangle-group method associates a finite-index subgroup of the triangle group of signature $(13, 3, 3)$ to a permutation triple. Expansions of holomorphic differentials give a canonical model of the source curve. For the chosen cover the canonical model is a genus-six curve in $\mathbb{P}^5$, realized as a quadric section of a smooth del Pezzo surface of degree 5. In particular, it is neither hyperelliptic, trigonal, nor a plane quintic. The canonical hyperplane divisor H has degree 10.

A marking of the del Pezzo surface is additional arithmetic data. The descent computation identified the Galois action on its five degree-four pencils with the full group S_5 . Coordinates adapted to an ordered marking therefore need not be rational even when the curve and covering map descend to $\mathbb{Q}$. Invariant data and the branch fibers produced an exact rational model, after which the map was represented by a ratio of sections of $\mathcal{O}_Z(8H)$. Some entries of the exact kernel vectors defining those sections have 480038 decimal digits. This size motivated a different generator of the function field rather than further manipulation of the same coordinates.

7.3 Generators with poles over infinity

The pole fiber is $13(P_0 + P_4)$, with the rational point P_0 and degree-four point P_4 identified in [Section 4.1](#). In the discovery model the first non-gap at P_0 is 7, so P_0 is not a Weierstrass point. Choose a nonconstant function $x \in L(7P_0)$. The coefficient bound is [Lemma 6.1](#).

For $x \in L(7P_0)$ the list consists of thirteen entries $7/13$ and fifty-two zeros, recovering $\deg_t c_i \leq \lfloor 7 \min(i, 13)/13 \rfloor \leq 7$. The function x generates $\mathbb{Q}(Z)/\mathbb{Q}(t)$ because the permutation action is primitive: a proper intermediate field would give a block system, while a function with only the pole P_0 cannot belong to $\mathbb{Q}(t)$. Trace-centering kills the coefficient of X^{64} . The resulting f has primitive integer height 249 digits. Its rational coefficients have numerators of at most 244 digits and common denominator 13^{182} ; these are separate coefficient statistics.

The same search in $L(3P_0 + P_4)$ produced F_3 . The pole bound and exact bridge are given in [Section 6](#).

7.4 Recovering the linear series without eliminating the base locus

Write $t = N/D$ with $N, D \in V = H^0(Z, \mathcal{O}_Z(8H))$, so $\dim V = 75$. Their common base divisor R has degree 15, and $\text{div}(D) = 13(P_0 + P_4) + R$. The condition that A/D belong to $L(7P_0)$ consists of vanishing of A along R , to order 13 along P_4 , and to order 6 at P_0 .

The base-divisor condition can be imposed by multiplication of sections. Indeed,

$$NV + DV = H^0(Z, \mathcal{O}_Z(16H)(-R)). \quad (12)$$

The right-hand side has dimension $145 + 1 - 6 = 140$. To compute the dimension on the left, remove the common divisor from N and D . The resulting pencil is base-point free. Its relations identify $NV \cap DV$ with $H^0(Z, \mathcal{O}_Z(R))$, which has dimension $15 + 1 - 6 = 10$. Hence the left-hand side also has dimension $150 - 10 = 140$, proving (12). Since V is globally generated,

$$A \text{ vanishes along } R \iff AV \subseteq NV + DV. \quad (13)$$

For the reverse implication, choose locally a section in V that is a unit along each point of R and apply (12).

This replaces explicit equations for the base locus by a linear-algebra test. Sections of $\mathcal{O}_Z(16H)$ are determined by jets of length 34 at the five geometric pole points: a section vanishing on all those jets would have at least 170 zeros, exceeding its degree 160. The same jets therefore suffice to implement (13).

The Riemann–Roch calculation was performed at 92 primes used for Chinese remaindering and at eight held-out primes. At every prime the computed space $L(7P_0)$ had dimension two and pole-order flag $(7, 0)$; all 473 coefficient slots were reconstructed, and an exact rational control identified the result with an affine change of f . Thus the calculation also supplies the stated first non-gap. The same run for $L(3P_0 + P_4)$ gave dimension two, pole-order flag $(3, 0)$, and all 319 slots reconstructed with the same held-out checks. The resultant calculation in [Section 6.2](#) then proves over $\mathbb{Q}(t)$ that its reduced polynomial has the same function field as f .

Choosing these generators turns the large section representation into a compact plane equation. The resulting singularities account for the apparent branch points treated in [Section 3](#).

8 Reproducibility and the proof package

The paper, exact data, and verification programs are available at <https://github.com/sebastianbiallas/sz8-over-q>. The exact coefficients of f are the starting input to the proof package. The programs verify the assertions listed below, including the changes of generator and both specializations. The canonical-model and modular reconstruction engines used in discovery are not included. In particular, the reconstruction controls in Section 7 do not establish an exact characteristic-zero identification of the upstream numerical curve with the canonical discovery model; the polynomial certificate supplies the proof of the stated realizations independently of that identification.

Polynomial	Role
$f(X, t), g(X, s)$	The certified auxiliary equation and its cleared pullback along $t = T(s)$
$F_3(Y, U), G_3(Y, s)$	The smaller equation, with $U = 3(t + 1)$, and its cleared pullback
$h(X), P(X)$	The fiber $f(X, -7/5)$ at $s = 0$ and a reduced integral equation for the same field
$P_1(X)$	The printed integral defining polynomial for the fiber $F_3(Y, 4)$ at $s = 1$

Table 1: Polynomial notation and roles in the proof package.

8.1 Verification stages and software

Stage	Mathematical assertion
Exact algebra	Discriminant identity, squarefreeness, branch-fiber multiplicities
Exact modular node check	Degree-one gcfs at every root of S ; the local lemma gives unramified normalization
Certified continuation	Two explicit permutations generate the geometric group
Finite groups	Group identification, self-normalizer, inertia normalizers
Cubic arithmetic	Identification of the rational quotient and the base change
Specialization	Full $Sz(8)$ group at $t = -7/5$ and $t = 1/3$
Changes of generator	Exact resultant bridge and printed number-field polynomial

Exact arithmetic uses FLINT through python-flint [FLI, pyt] and PARI/GP [PAR]; finite-group computations use GAP [GAP]; ball computations use Arb [Joh17] through python-flint; permutation-group computations also use SymPy [M⁺17]. For the recorded run, the discriminant and parameter-root stage took about 488 seconds, the node stage about 7 seconds, and the two monodromy loops together about 1518 seconds on the source machine. These timings describe the original interval-based run. The modular node check took about 44 seconds on a different machine, including discriminant interpolation modulo 1000003. The full-run driver uses the interval proof; the modular check is a separate exact alternative, conditional on the characteristic-zero discriminant identity (3).

The accompanying README gives separate commands for a short exact-data check and a full certificate run. The short check verifies file digests, the cubic pullback, specialization factorizations, and the group generated by the stored permutations; establishing those permutations as monodromy requires the certified continuation. SHA-256 hashes tie reports to their exact inputs.

8.2 Path and enclosure parameters

For the node test, parameter centers are exact decimal rationals and radii are integer powers of 10. The consumer re-certifies each parameter rectangle at its own working precision before evaluating f on it. All complex rectangles are enclosed by Euclidean discs using their half-diagonals.

Each meridian uses a 24-sided polygon inscribed in a circle of radius $\varepsilon \simeq 0.00151183882584$. The reported minimum distance of the radial segments from node centers is about 0.00453551647752. These decimals describe the path geometry; the continuation tests apply to the actual segments between the stored-precision waypoints.

For each root enclosure with center c_i and radius r_i , the program tries $R_i = \alpha d_i$, where $d_i = \min_{\ell \neq i} |c_i - c_\ell|$ and $\alpha \in \{0.1, 0.03, 0.01, 0.003, 0.001\}$. It retains only radii greater than $4r_i$ and chooses the one allowing the largest certified step under the Rouché bounds in [Section 3](#). If derivative-root isolation fails, the fallback bounds the derivative term by term in t , using precomputed factorizations of the coefficient polynomials in X . Both meridians were run at 1200 bits and required 1564 accepted steps each.

8.3 Data formats and exact inputs

The file `data/f.json` specifies the 472 nonzero terms of f exactly. An entry with key `"i,j"` and rational value `"a/b"` denotes the term $(a/b)X^i t^j$; omitted terms are zero. In particular, the coefficient of X^{65} is 1, and that of X^{64} is 0.

Its SHA-256 digest is `4f88390af6e2119988e252d5ea62629e25c8a8bbe79bb63e56adddf18e0f319e`.

The file `data/g.json` gives the cleared-denominator polynomial (9), of primitive integer height 256 digits. A coefficient list for key `"i"` gives the coefficient of X^i in ascending powers of s . The file `data/monodromy.json` records σ_1 and σ_2 as zero-based image lists: entry i is the image of letter i . Their product means apply σ_1 first, then σ_2 .

The directory `certificate/` contains verification programs, recorded reports, a finite-group script, and a losslessly compressed exact discriminant artifact. The latter also contains parameter-root enclosures and can be regenerated from f . A manifest records the digests of all certificate files.

These files mark the reproducible boundary of the argument. The proof of the realizations uses the exact polynomial data, the algebraic checks, and the certified continuation described here. The discovery model explains how those inputs were found, but is not a trusted input to the proof.

A The polynomial P_1

The following table gives every coefficient of

$$P_1(X) = \sum_{i=0}^{65} a_i X^i \in \mathbb{Z}[X].$$

This monic irreducible polynomial defines the $s = 1$ specialization field of [Corollary 1.2](#). Its Galois group over $\mathbb{Q}$ is $\text{Sz}(8)$, its signature is $(1, 32)$, and its largest coefficient has 43 decimal digits. The exact field isomorphism and the two Frobenius witnesses are described in [Section 6.3](#).

i	a_i	i	a_i
65	1	32	111576271656374943649168627144
64	-26	31	-1830375164331879075601081424722
63	286	30	14165048453505707875648525322302
62	-3042	29	-84019769012884129710271324364208
61	48516	28	414402298181737353455412295898476
60	-613210	27	-1711899045061481063649279772181022
59	5908474	26	5626339602414031754039539622828196
58	-58994624	25	-11503816871202076427393433090432024
57	623518571	24	-17315770588309753701747702884908822
56	-6115614518	23	355731138138350959453558761085591892
55	57704935210	22	-2477607217524933423614777734799019642
54	-540988565624	21	12952487803552222023226153351026496120
53	4930801652920	20	-57298733689275698346414841298028074074
52	-43837618902168	19	223604728219527065726389644847612608884
51	379354666262442	18	-784015815015555570393621131248127983622
50	-3159436411932060	17	2492316653651004236699613636872505693797
49	25554067710938338	16	-7212668264254213241823125912501947296476
48	-201830975114372136	15	19034588421757576035880412511758188395692
47	1549057447659288814	14	-45795161530711524477456818311388615520734
46	-11640105719955601434	13	100304804988436052362644640445362952553490
45	86347446925333503156	12	-199386324066139251618199918828464163139310
44	-627105865931437120956	11	358043144182771691993942421181392092476748
43	4412516453638598194656	10	-576709148417879295069158359422560844254702
42	-29998300854797520006060	9	824896024833023465152535113896403007200031
41	196423272421894760745204	8	-1027388981955420848083824140671306994720518
40	-1229545463807424121488624	7	1079821738731656528124771338204008334304456
39	7299785566669735636659772	6	-893542087860684403720186773246458268957364
38	-40809292572830325238266962	5	454221033119485870609889964702713666035710
37	212828899967877893394852612	4	110949725422052183718013061296077304350108
36	-1018475736998960925560944926	3	-561998186642565191556265162501607480469848
35	4324985869023705716185739514	2	735581423524295925619570416058856384826452
34	-14990135260656792314064163216	1	-584358399988542382058842060050072760557946
33	29778317591940060578619748499	0	232507129296403674822330322023465543363816

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